VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering

Project Report on

KrishiMitra – AI-Powered Smart Agriculture Assistant

In partial fulfillment of the Fourth Year (Semester–VII), Bachelor of Engineering (B.E.) Degree in Computer Engineering at the University of Mumbai

Academic Year 2025-26

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(2025-26)

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CERTIFICATE of Approval

This is to certify that **Jenny Lalwani 37(D17A), Sahil Ahuja 03(D17A),Vineet Chelani 17(D17A) ,Sahil Ahuja 03(D17B)** of Fourth Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on "***title of the project***" as a part of the coursework of PROJECT-I for Semester-VII under the guidance of ***xyx.*** in the year 2024-2025.

_____

Date

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Computer Engineering Department

COURSE OUTCOMES FOR B.E PROJECT

Learners will be to:-

Course Outcome	Description of the Course Outcome
CO 1	Do literature survey/industrial visit and identify the problem of the selected project topic.
CO2	Apply basic engineering fundamental in the domain of practical applications FORproblem identification, formulation and solution
CO 3	Attempt & Design a problem solution in a right approach to complex problems
CO 4	Cultivate the habit of working in a team
CO 5	Correlate the theoretical and experimental/simulations results and draw the proper inferences
CO 6	Demonstrate the knowledge, skills and attitudes of a professional engineer & Prepare report as per the standard guidelines.

ABSTRACT

The advancement of Artificial Intelligence (AI) and Machine Learning (ML) has opened new opportunities in the agricultural sector, helping farmers make informed, data-driven decisions. The KrishiMitra – AI-Powered Agricultural Assistant is a smart agriculture platform designed to empower farmers with personalized insights, predictive analytics, and intelligent automation. The system integrates multiple AI-based modules such as Crop Recommendation, Plant Disease Detection, AI-Based Weekly Crop Plan Generator, RAG-Based Chatbot, and Market Analysis, creating a unified ecosystem that addresses various agricultural challenges.

The Crop Recommendation System utilizes machine learning algorithms like Random Forest to suggest the most suitable crops based on soil type, climate, and regional conditions. The Plant Disease Detection module employs a Convolutional Neural Network (CNN) to identify crop diseases from uploaded images and provides appropriate treatment suggestions. The AI-Based Weekly Plan Generator dynamically creates week-wise farming schedules covering seed selection, irrigation, fertilization, pest control, and harvesting recommendations. The RAG-Based Chatbot, powered by Sentence Transformers, FAISS vector storage, and the Mistral-7B LLM, enables farmers to ask agricultural queries in natural language and receive accurate, contextual responses. Additionally, the Market Analysis Module fetches real-time crop price and demand data, helping farmers make better economic decisions and connect directly with buyers.

The system is developed using modern web technologies such as React.js and Tailwind CSS for the frontend, Flask and Node.js for the backend, and MongoDB/PostgreSQL for data storage. This integrated approach ensures a scalable, secure, and user-friendly experience. Overall, KrishiMitra aims to bridge the gap between technology and agriculture, promoting sustainable farming, reducing crop losses, and enhancing productivity through intelligent, accessible, and multilingual AI solutions

INDEX

Abstract **5**
Chapter 1 Introduction **7**
1.1 Introduction to the Project 7
1.2 Motivation 8
1.3 Problem Definition 8
1.4 Relevance of the Project 8
1.5 Methodology 9
Chapter 2 Literature Survey **10**
2.1 Research Papers Referred 10
2.2Books / Journals / Articles Referred 12
2.3Interactions with Domain Experts 13
2.4Patent Search 13
2.5Existing Systems 15
2.6Lacuna in the Existing Systems 16
2.7Comparison of Existing Systems and Proposed Area of Work 17
2.8Focus Area 18
Chapter 3 Requirements **19**
3.1 Proposed Model 19
3.2 Functional Requirements 19
3.3 Non-Functional Requirements 20
3.4 Hardware and Software Requirements 20
3.5 Technology and Tools Utilized 21
3.6 Constraints of Working 21
Chapter 4 Proposed Design **22**
4.1 Block Diagram of the proposed system 22
4.2 Detailed Design 23
4.3 Proposed Algorithms 24
4.4 Project Scheduling / Gantt Chart 25
Chapter 5 Results and Discussions **26**
5.1 Crop Recommendation 26
5.2 Plant Disease Accuracy 27
5.3 Market Analysis 27
5.4 AI Agricultural Assistant 28
Chapter 6 Plan Of Action For the Next Semester **29**
6.1 Work done till date 29
6.2 Plan of action for project II 29
Chapter 7 Conclusion **30**
References **31**
Appendix **31**
A. List of Figures 32
B. List of Tables 32
C. Review 1 Sheet 32

Chapter 1 : Introduction

This chapter introduces the *KrishiMitra – AI-Powered Agricultural Assistant* project and outlines its overall concept and objectives. It explains the motivation behind developing the system, defines the key problems addressed, and highlights the project's significance in modern agriculture. The chapter also provides an overview of the methodology and technologies used to implement the proposed solution.

1.1 Introduction

Agriculture forms the backbone of the Indian economy, providing livelihoods for a large portion of the population. However, farmers often face numerous challenges such as unpredictable weather, pest infestations, poor market access, and lack of real-time advisory support. With the advent of Artificial Intelligence (AI) and data-driven systems, modern technologies can now be leveraged to overcome these traditional challenges.KrishiMitra is designed as a comprehensive digital platform that empowers farmers with intelligent decision-making tools. It provides crop recommendations based on soil and environmental data, predicts plant diseases through image analysis, generates AI-based weekly crop plans, and offers market insights and chatbot-based agricultural guidance. The platform integrates machine learning, computer vision, and natural language processing to create a one-stop solution for precision farming and sustainable agriculture.

1.2 Motivation

The primary motivation behind developing KrishiMitra is the growing need to modernize farming practices and bridge the information gap faced by farmers in rural areas. Many farmers still rely on traditional methods, lacking access to scientific insights or expert consultation. Crop failures due to diseases, wrong crop selection, or market misjudgment lead to economic losses and lower productivity.

The project aims to use AI-driven insights to assist farmers in making data-backed decisions at every stage of cultivation—from seed selection to marketing. By providing localized,

personalized, and multilingual support through an intelligent chatbot and analytical modules, KrishiMitra aspires to make agriculture more efficient, profitable, and sustainable.

1.3 Problem Definition

Farmers today face numerous challenges in agriculture due to the lack of real-time, data-driven guidance and technological support. Traditional farming practices often lead to inefficient crop selection, delayed disease detection, poor resource management, and limited market access. Most farmers rely on experience or local advice rather than scientific insights, which increases the risk of crop failure and reduces profitability. There is also an absence of an integrated system that combines crop recommendation, disease diagnosis, market analysis, and personalized guidance in one platform. To address these issues, the project *KrishiMitra – AI-Powered Agricultural Assistant* aims to develop an intelligent, all-in-one platform that leverages artificial intelligence to assist farmers in every stage of cultivation. By integrating machine learning for crop recommendation, computer vision for disease detection, and a RAG-based chatbot for agricultural advisory, KrishiMitra seeks to bridge the gap between technology and farming, empowering farmers with timely, accurate, and localized decision support.

1.4 Relevance of the Project

The project is highly relevant in today's era of smart and sustainable farming. It bridges the gap between traditional agricultural practices and modern AI-driven technologies, helping farmers make data-backed decisions. By providing crop recommendations, disease detection, and real-time market insights, it enhances productivity and profitability. The system aligns with India's Digital Agriculture Mission and the UN Sustainable Development Goal of promoting sustainable agriculture. KrishiMitra empowers farmers with personalized, multilingual, and region-specific guidance through an intelligent chatbot interface. Its predictive and analytical features reduce risks, optimize resource use, and minimize dependency on manual expertise. Overall, the project contributes to transforming agriculture into a more efficient, environmentally conscious, and economically viable sector.

1.5 Methodology Used

KrishiMitra's system is built using a modular AI architecture that integrates data processing, model training, and user interaction components. The methodology can be summarized as follows:

1. **Data Collection & Preprocessing:**
    - Soil type, climate, and regional datasets are gathered for crop recommendation.
    - Plant image datasets are used for disease classification using CNNs.
    - Agricultural text data is indexed using FAISS for chatbot retrieval.

2. **Machine Learning & AI Implementation:**
    - **Crop Recommendation:** Implemented using supervised ML models that analyze soil and environmental parameters.
    - **Disease Detection:** CNN models (via OpenCV) identify crop diseases from uploaded images.
    - **Chatbot Module:** RAG-based system using Mistral-7B LLM, Sentence Transformers for embeddings, and FAISS for vector search.

3. **Backend Development:**
    - Flask handles the ML models and inference APIs.
    - Node.js (Express.js) manages chatbot and market-related endpoints.

4. **Frontend Development:**
    - React.js with Tailwind CSS provides an intuitive, multilingual interface for farmers.

5. **Database & Storage:**
    - MongoDB/PostgreSQL store structured crop, user, and market data.
    - FAISS handles embedding-based search for chatbot responses.

Chapter 2: Literature Survey

This chapter reviews relevant research and background material that informs the KrishiMitra project. It summarizes the core findings and contributions of key research papers, lists useful books/journals/articles, reports insights collected from domain interactions, provides a focused patent study, describes existing systems in the market, identifies gaps in those systems, and finally compares existing solutions to the proposed KrishiMitra approach while highlighting the primary focus areas.

2.1 Research Papers Referred

1. A. Vijayvargia et al., "Intent Aware Context Retrieval for Multi-Turn Agricultural Question Answering," AIDE 2025

Abstract:
This paper introduces a context-aware retrieval framework for agricultural question answering systems. It focuses on intent recognition and context preservation in multi-turn conversations, improving how chatbots understand farmers' sequential queries. The model uses a hybrid retrieval approach combining semantic similarity and intent classification to fetch accurate answers, even when users use local or unstructured language.It highlights how agricultural assistants can become more interactive, adaptive, and personalized by understanding conversational flow and intent.

Inference:
For KrishiMitra, this approach strengthens the chatbot module. By integrating intent-aware retrieval, the system can provide contextually relevant agricultural advice across multiple turns of conversation, enhancing the reliability and natural flow of farmer–AI interaction.

2. S. Gupta et al., "AI-Driven Crop Recommendation System Using Soil and Climate Data," IEEE Access, 2024

Abstract:
The paper presents a machine learning-based framework that recommends optimal crops based

on soil parameters such as pH, nitrogen, phosphorus, potassium, and climatic conditions like temperature and rainfall. The Random Forest and Gradient Boosting models achieved higher accuracy compared to traditional regression models. The system aims to maximize yield by aligning environmental factors with suitable crops.It emphasizes the potential of AI in precision agriculture, helping farmers choose the best crops for sustainable production.

Inference:

KrishiMitra's crop recommendation module is built upon similar principles. By adapting such AI models, it can provide region-specific crop suggestions, improving decision-making accuracy and resource utilization while boosting productivity.

3. N. Kumar et al., "Plant Disease Detection Using Deep Learning Techniques," Computers and Electronics in Agriculture, 2023

Abstract:

This study explores the use of Convolutional Neural Networks (CNNs) for classifying plant diseases from leaf images. It compares various architectures such as VGG16, ResNet50, and MobileNet, showing that deep learning significantly outperforms traditional image processing techniques. The research also emphasizes dataset diversity, augmentation, and real-time deployment feasibility.The approach demonstrates the effectiveness of computer vision in early disease diagnosis, reducing crop loss through timely intervention.

Inference:

The CNN-based disease detection module in KrishiMitra is inspired by this concept. Implementing lightweight architectures like MobileNet ensures faster inference on mobile devices, making the system more practical for farmers in remote areas.

4. P. Singh and M. Patel, "AgriMarket: AI-Based Agricultural Market Analysis and Price Forecasting," Elsevier Journal of Smart Systems, 2024

Abstract:

This paper proposes a price forecasting model for agricultural commodities using time-series analysis and machine learning. The authors utilize LSTM (Long Short-Term Memory) networks to predict market price trends based on historical data, rainfall, and crop output. The study aims

to minimize farmer losses caused by fluctuating market prices and lack of information.The system enables farmers to make data-driven selling decisions and choose profitable marketplaces.

Inference:
KrishiMitra's market analysis feature can incorporate a similar LSTM-based forecasting model to predict future price trends, helping farmers optimize sales timing and improve profitability.

5. R. Sharma et al., "A Comprehensive Review of Smart Farming and IoT-Based Agriculture Systems," Springer Nature, 2024

Abstract:
The paper reviews advancements in IoT-enabled smart farming systems and their integration with Artificial Intelligence. It discusses data collection from soil sensors, drones, and weather APIs, combined with predictive analytics for precision farming. The study also highlights challenges in data interoperability, scalability, and farmer adoption of AI-driven tools.It concludes that hybrid AI-IoT systems can significantly enhance agricultural efficiency and sustainability.

Inference:
KrishiMitra aligns closely with this vision by integrating AI, data analytics, and real-time insights. It has the potential to evolve into a complete AgriTech ecosystem, connecting IoT-based data with intelligent decision support to modernize farming practices.

2.2 Books / Journals / Articles Referred

Several books, journals, and articles were referred to for developing a comprehensive understanding of Artificial Intelligence and its applications in agriculture. The book *"Artificial Intelligence in Agriculture: Emerging Applications and Technologies"* by P. Singh provided valuable insights into how AI models, data analytics, and IoT devices are revolutionizing farming practices. Articles from *IEEE Transactions on Artificial Intelligence in Agriculture* and *Elsevier's Smart Agricultural Systems Journal* offered detailed case studies on crop recommendation systems, plant disease detection using CNNs, and market forecasting with

time-series models. Additionally, journals such as *Agricultural Informatics Review* and *Computers and Electronics in Agriculture* were instrumental in understanding how machine learning and vision-based models improve precision farming. These resources collectively guided the conceptual framework of *KrishiMitra* by offering technical direction for building efficient, scalable, and farmer-friendly AI-powered systems.

2.3 Interactions with Domain Experts

- **Agronomist:** Emphasized the need for localized recommendations (micro-climates, seasonal calendars) and the importance of validating disease-symptom mappings with field trials rather than only lab images. Advised inclusion of explanatory guidance (why a recommendation is made and possible alternatives).
- **Extension Officer:** Highlighted farmer constraints limited smartphone literacy, intermittent connectivity and recommended a simple UI, regional languages, SMS/IVR fallbacks, and offline-first functionality.
- **AI Researcher / ML Engineer:** Recommended lightweight, quantized models for on-device inference, and stressed robust data augmentation and cross-domain validation to handle field-image variability.

2.4 Patent Search

2.4.1. US10950123B2: System and Method for Automated Crop Disease Detection Using Machine Learning (2021)

Abstract:

This patent presents a method that leverages machine learning for the automated detection of crop diseases through image-based analysis. It involves capturing plant images, performing preprocessing to remove noise, extracting key features using convolutional neural networks (CNNs), and classifying them into healthy or diseased categories. The system can be deployed using mobile devices or IoT cameras for on-field monitoring.The patent demonstrates how CNNs can effectively detect crop diseases but focuses primarily on disease identification. In *KrishiMitra*, this concept is extended by integrating a treatment recommendation engine and

adaptive learning based on environmental and crop conditions.

US Patent Link: https://patft.uspto.gov/patft/search/

2.4.2. EP3689124A1: Smart Agricultural Advisory System Based on Soil and Weather Data (2020)

Abstract:

This European patent describes an expert advisory system that uses soil composition, temperature, humidity, and rainfall data to recommend suitable crops and irrigation plans. The system collects data from sensors and applies rule-based algorithms to provide recommendations for improving yield. While effective in static recommendations, the system lacks the adaptability of AI-based models. *KrishiMitra* enhances this approach by employing machine learning models that dynamically adjust recommendations based on seasonal and regional variations.

European Patent Link: https://worldwide.espacenet.com/patent/search/family/0689124

2.4.3. US20210234567A1: AI Chatbot for Farmer Assistance (2021)

Abstract:

This patent introduces a conversational AI chatbot designed to assist farmers by answering queries related to crop management, soil quality, and pest control. It uses NLP techniques for understanding queries and retrieving predefined responses from a database. Although it provides basic conversational support, it lacks contextual retrieval and reasoning capabilities. *KrishiMitra* upgrades this concept using a Retrieval-Augmented Generation (RAG)-based multilingual chatbot powered by FAISS and Mistral-7B to deliver intelligent, localized, and context-aware responses.

US Patent Link: https://patft.uspto.gov/patft/search/

2.4.4. EP3746210A1: Market Analysis and Forecasting System for Agricultural Produce (2021)

Abstract:

This invention focuses on analyzing market trends and predicting demand for agricultural commodities using real-time pricing data. The system employs time-series forecasting and regression models to help farmers decide optimal selling times.

Inference:

The patent effectively handles market forecasting but does not integrate with broader farm management systems. *KrishiMitra* expands this by linking market intelligence with crop planning, disease management, and direct produce selling, creating a unified AI ecosystem for farmers.

European Patent Link: https://worldwide.espacenet.com/patent/search/family/0746210

2.5 Existing Systems

1. **CropIn SmartFarm**
    - A digital agriculture platform that offers farm management, weather insights, and yield prediction using satellite imagery and AI.
    - Mainly used by agribusinesses rather than individual farmers.
    - Lacks personalized recommendations or interactive advisory features.
2. **Plantix**
    - A mobile-based application that helps farmers identify plant diseases by uploading crop images.
    - Uses image processing and machine learning for disease classification and provides basic treatment advice.
    - Does not integrate crop planning, market analysis, or adaptive learning modules.
3. **Kisan Suvidha (Government of India)**
    - Provides farmers with information about weather forecasts, market prices, and farming advisories.
    - Focuses on data dissemination but lacks AI-driven analytics or personalized decision support.
4. **Farmonaut**
    - Offers satellite-based farm monitoring and soil analysis tools for efficient crop management.
    - Effective for precision farming but does not include AI chatbots or real-time farmer interaction systems.
5. **AgroAI**
    - Uses machine learning for crop recommendation and pest detection.

- Operates as an academic model with limited scalability and lacks integration with real-time market data or multilingual farmer communication.

6. **Overall Comparison**
    - Most existing systems are domain-specific, focusing only on one aspect such as crop health, pricing, or weather prediction.
    - None provide a unified AI-powered platform combining crop recommendation, disease detection, chatbot support, and market insights.
    - *KrishiMitra* bridges this gap by integrating all these features into one comprehensive, farmer-centric solution.

2.6 Lacuna in the Existing Systems

1. **Lack of Integration:** Existing systems address specific areas like crop recommendation or disease detection but do not offer an all-in-one intelligent platform for complete farm management.
2. **Limited Personalization:** Most tools provide static, rule-based advice without adapting to changing weather, soil, or regional conditions, leading to less effective recommendations.
3. **Absence of Intelligent Crop Planning:** Current systems do not generate dynamic, week-wise crop plans based on real-time data and crop growth stages.
4. **No Interactive or Multilingual Support:** Farmers lack access to smart AI chatbots that can provide region-specific assistance in local languages.
5. **Weak Market Connectivity:** Few platforms connect farmers directly to buyers or provide real-time market analytics, reducing their profit opportunities.

2.7 Comparison of Existing Systems and Proposed Area of Work

Feature	Existing Systems	Proposed System – KrishiMitra
Integration of Modules	Most existing tools focus on a single functionality such as crop recommendation, disease detection, or market forecasting.	Combines multiple AI-powered features—crop recommendation, disease prediction, crop planning, chatbot, and market analysis—into one unified platform.
Personalization	Provides static or generalized advice without adapting to local soil, climate, or seasonal variations.	Offers personalized, data-driven recommendations using real-time inputs and machine learning models.
Technology Used	Primarily rule-based systems or basic ML models with limited scalability.	Employs deep learning (CNN, Transformers), RAG-based LLM chatbot, and FAISS vector database for efficient and scalable AI operations.
User Interaction	Minimal interaction, often limited to basic dashboards or one-way advisories.	Interactive multilingual AI chatbot powered by Mistral-7B that enables conversational assistance in regional languages.

Crop Planning	Absent or static; users rely on manual scheduling.	Dynamic week-wise crop plan generator adapting to crop growth stage, weather, and soil conditions.
Market Connectivity	Provides only market price data, with no direct linkage to buyers.	Integrates market analytics and direct selling options to help farmers maximize profits.
Accessibility	Limited availability in rural areas and requires technical knowledge.	Simple web-based interface designed for ease of use by farmers with minimal technical expertise.

Table 1. Comparison of existing systems and proposed area of work

2.8 Focus Area

1. The main focus area of this study is the integration of **IoT and AI technologies** to develop a smart and efficient agricultural system that assists farmers in real-time decision-making.
2. It emphasizes **data collection through IoT sensors** for monitoring soil, crop, and environmental parameters to improve productivity and sustainability.
3. The project focuses on **AI-driven analysis and prediction**, including crop recommendation, disease detection, and market forecasting.
4. Another key area is the development of a **user-friendly interface** that provides personalized insights and actionable recommendations for farmers.
5. The study also aims to explore **scalability, cost-effectiveness, and automation** in agricultural solutions to make technology accessible and beneficial for all types of farmers.

Chapter 3: Requirements

This chapter describes the key requirements needed to develop the KrishiMitra – AI-Powered Agricultural Assistant. It highlights the system model, functional and non-functional needs, hardware/software setup, tools used, and the main working constraints to ensure effective system implementation.

3.1 Proposed Model

The proposed **KrishiMitra** model integrates modern technologies such as **IoT, Artificial Intelligence, and Machine Learning** to assist farmers in making data-driven agricultural decisions. The system collects real-time environmental and soil data through sensors, analyzes it using AI models, and provides actionable insights. It recommends suitable crops, detects plant diseases, and offers AI-powered chatbot support for agricultural queries. Additionally, it incorporates market analysis to help farmers stay informed about crop demand and pricing trends. The model aims to empower farmers with intelligent, accessible, and technology-driven solutions for sustainable agriculture.

3.2 Functional Requirements

The functional requirements define the essential operations and features that the system must perform.

1. **User Authentication:** Provides secure login and access control for users.
2. **IoT Data Collection:** Gathers real-time data on soil moisture, temperature, humidity, and pH levels.
3. **Crop Recommendation:** Suggests optimal crops based on soil and environmental conditions using AI models.
4. **Disease Prediction:** Detects plant diseases from leaf images using Convolutional Neural Networks (CNN).
5. **AI Chatbot Assistance:** Offers natural language-based query support using Mistral-7B and RAG techniques.

6. **Market Analysis:** Displays market demand, pricing trends, and yield estimates for selected crops.
7. **Dashboard Visualization:** Provides interactive graphs and data insights for easy understanding.
8. **Alerts & Notifications:** Sends alerts regarding weather, irrigation schedules, or disease risks.

3.3 Non-Functional Requirements

1. **Performance:** The system should process large volumes of data efficiently with minimal delay.
2. **Scalability:** Capable of handling multiple farms and users simultaneously without performance degradation.
3. **Reliability:** Ensures continuous data monitoring and robust functioning of IoT devices.
4. **Usability:** Designed with a simple, intuitive interface accessible to users with minimal technical expertise.
5. **Security:** Implements strong authentication and data encryption for secure information exchange.
6. **Maintainability:** Built with modular architecture to allow easy updates and feature enhancements.
7. **Availability:** Should remain accessible with minimal downtime for continuous monitoring.

3.4 Hardware and Software Requirements

Hardware Requirements:

- **Processor:** Intel i5
- **RAM:** 8 GB
- **Storage:** 256 GB SSD
- **GPU:** NVIDIA CUDA-enabled GPU

Software Requirements:

- **Frontend:** React.js, Tailwind CSS
- **Backend:** Flask / Node.js
- **Database:** MongoDB / PostgreSQL
- **AI/ML Frameworks:** TensorFlow, PyTorch, Mistral-7B
- **Supporting Tools:** OpenCV, FAISS, Sentence Transformers
- **Cloud / Hosting:** AWS, Google Cloud, or Local Server

3.5 Technology and Tools Utilized

1. **Machine Learning Models:** CNN for disease prediction and AI-based crop recommendation.
2. **Flask & Node.js:** Backend frameworks for API handling and server-side logic.
3. **React.js & Tailwind CSS:** Used for creating an interactive and responsive user interface.
4. **FAISS & Sentence Transformers:** Enable efficient semantic search for chatbot intelligence.
5. **Mistral-7B Model:** Used in the chatbot for generating context-aware agricultural responses.
6. **OpenCV:** For image processing in disease detection modules.

3.6 Constraints of Working

1. **Internet Connectivity:** Limited or unstable connectivity in rural regions may affect real-time data transfer.
2. **Sensor Limitations:** Inaccurate readings due to environmental factors or hardware faults.
3. **Data Scarcity:** Lack of region-specific crop or disease datasets may limit AI accuracy.
4. **Hardware Costs:** Deployment of IoT devices and maintenance can be financially challenging.
5. **Power Supply Issues:** Continuous power availability is required for uninterrupted sensor operations.

Chapter 4: Proposed Design

This section outlines the design and workflow of the KrishiMitra system. It illustrates how various modules such as Crop Recommendation, Disease Prediction, Weekly Plan Generator, Chatbot, and Market Analysis interact to provide intelligent farming support. The design highlights the integration of machine learning models, LLM components, and a vector database to process data efficiently and deliver accurate, real-time recommendations to farmers.

4.1 Block Diagram of the proposed system

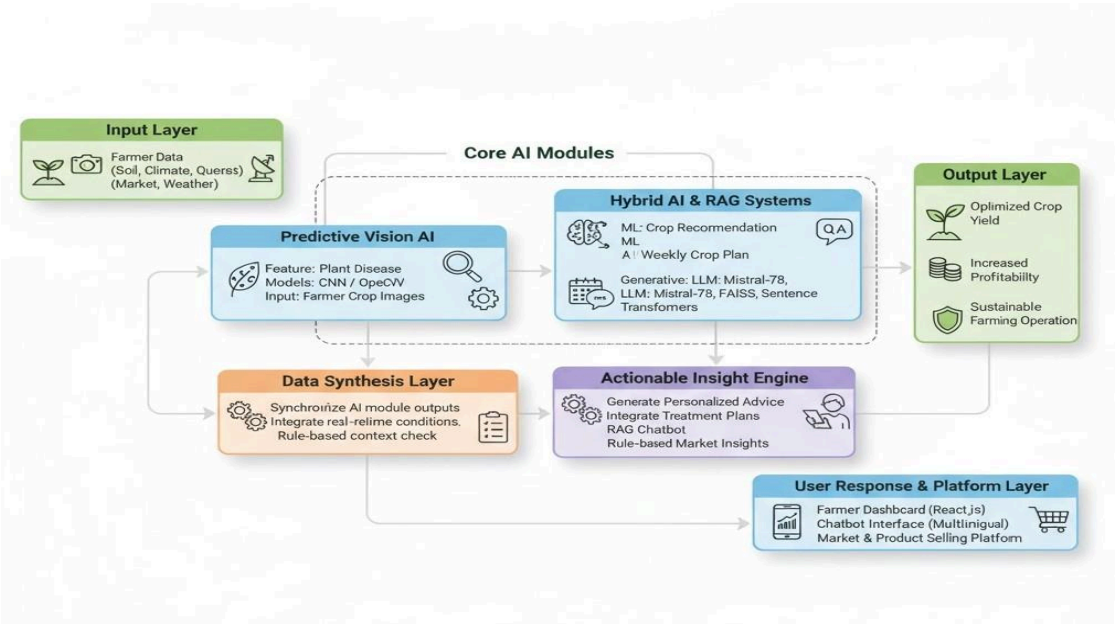


Fig 1: Block Diagram

Explanation of System Architecture

1. **User Interaction:**

   The system starts with user input, where the farmer interacts with modules like Crop Prediction, Disease Prediction, Weekly Plan Generator, Chatbot, and Market Analysis.

2. **Crop Prediction:**

   Users provide soil, region, and climate details, which are processed using a **Random Forest Model** to recommend the most suitable crops.

3. **Disease Prediction:**

   The user uploads a crop image, and the **CNN model** analyzes it to detect plant diseases and provide possible treatments.

4. **Weekly Plan Generator:**

   Based on user input, the **LLM (Gemini)** creates a dynamic, week-by-week crop management plan including sowing, irrigation, and harvesting steps.

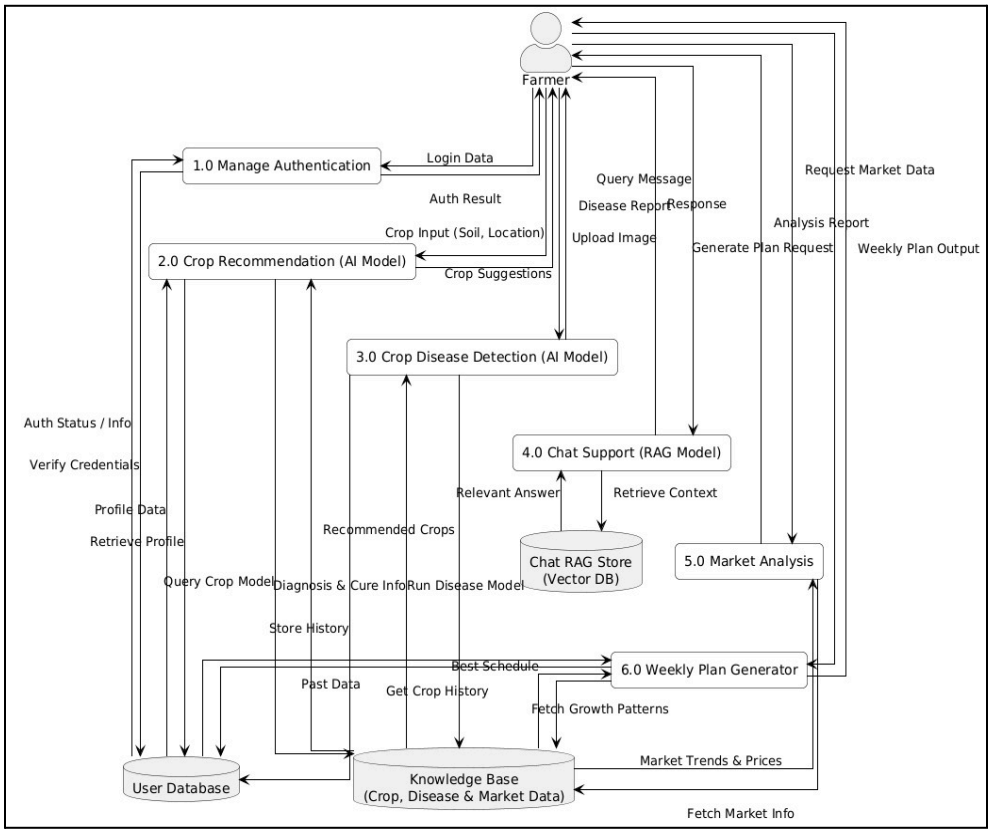
5. **Chatbot Module:**

   User queries are converted into **vector embeddings**, searched semantically in a **Vector Database (FAISS)**, and answered using an **LLM-based response system**.

6. **Market Analysis:**

   The system fetches real-time crop prices and market trends from **Data.gov.in**, helping farmers make informed selling and buying decisions.

4.2 Detailed Design



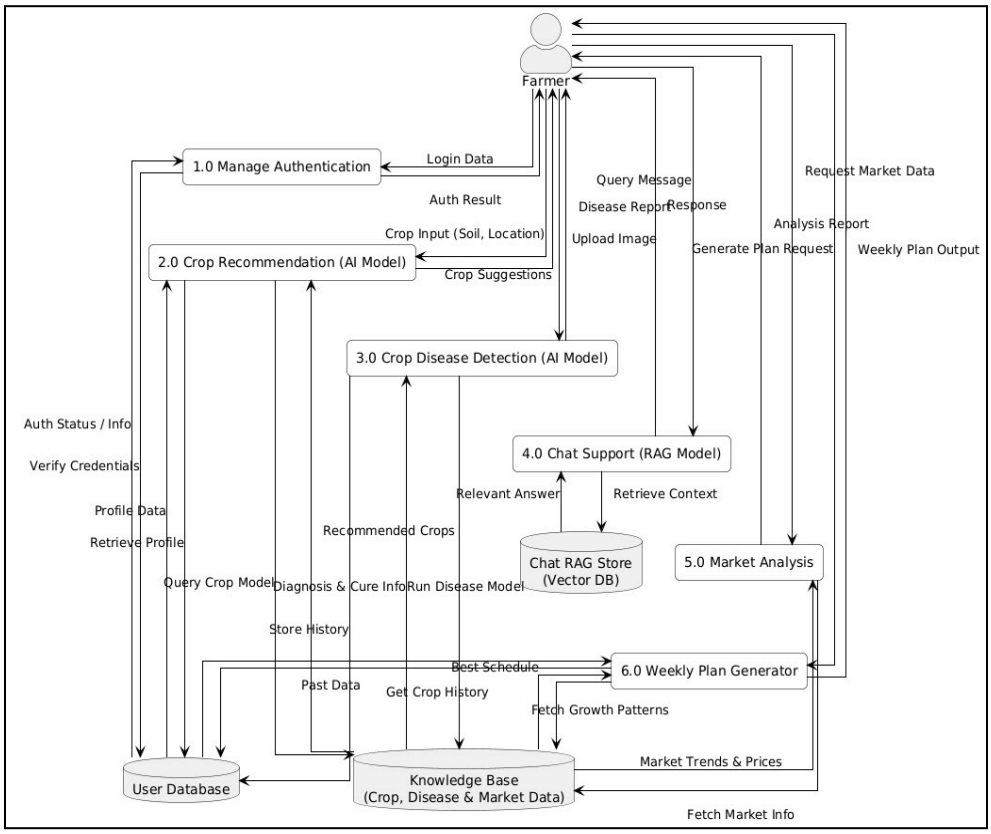


Fig 3: DFD (Level 1)

4.3 Proposed Algorithms

The KrishiMitra system utilizes a combination of machine learning and deep learning algorithms to deliver intelligent agricultural insights across its modules. Each algorithm is chosen for its efficiency, accuracy, and suitability for specific agricultural tasks.

1. **Crop Recommendation – Random Forest Algorithm**
    - The Random Forest model analyzes soil parameters, temperature, humidity, and rainfall to recommend the most suitable crops for cultivation.
    - It works by constructing multiple decision trees and combining their outputs to improve prediction accuracy and reduce overfitting.
2. **Plant Disease Detection – Convolutional Neural Network (CNN)**

- The CNN model is used for classifying crop images to identify diseases based on leaf color, texture, and pattern features.

3. **Chatbot and Query Response – Sentence Transformer + FAISS + LLM (Mistral-7B)**
    - User queries are converted into vector embeddings using **Sentence Transformers**.
    - These embeddings are searched through **FAISS**, a vector database, to retrieve relevant information.
    - The **Mistral-7B LLM** then generates context-aware and human-like responses in multiple languages.
4. **AI-Based Weekly Plan Generator – LLM (Gemini/Mistral)**
    - The LLM dynamically generates week-wise crop plans, including sowing, irrigation, and harvesting schedules.
    - It adapts recommendations based on crop type, stage, and real-time environmental data.
5. **Market Analysis – Regression and Trend Prediction Models**
    - Regression algorithms analyze real-time market datasets to predict price trends and demand variations.
    - This helps farmers make informed selling decisions and maximize profits.

4.4 Project Scheduling / Gantt Chart

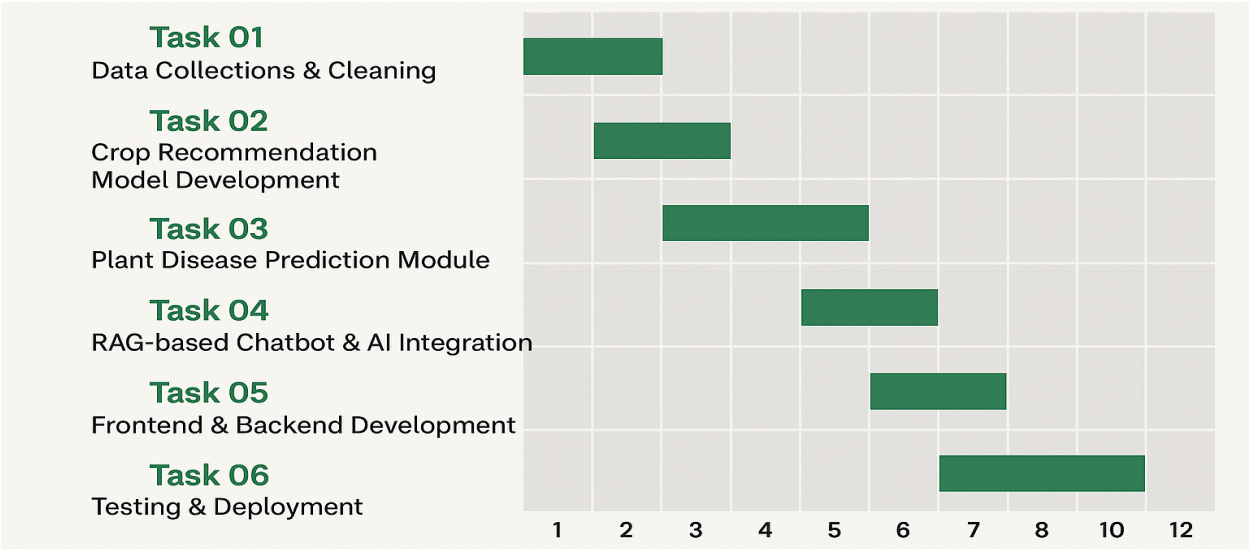


Fig 4 : Gantt Chart

Chapter 5. Results and Discussions

5.1 Crop Recommendation

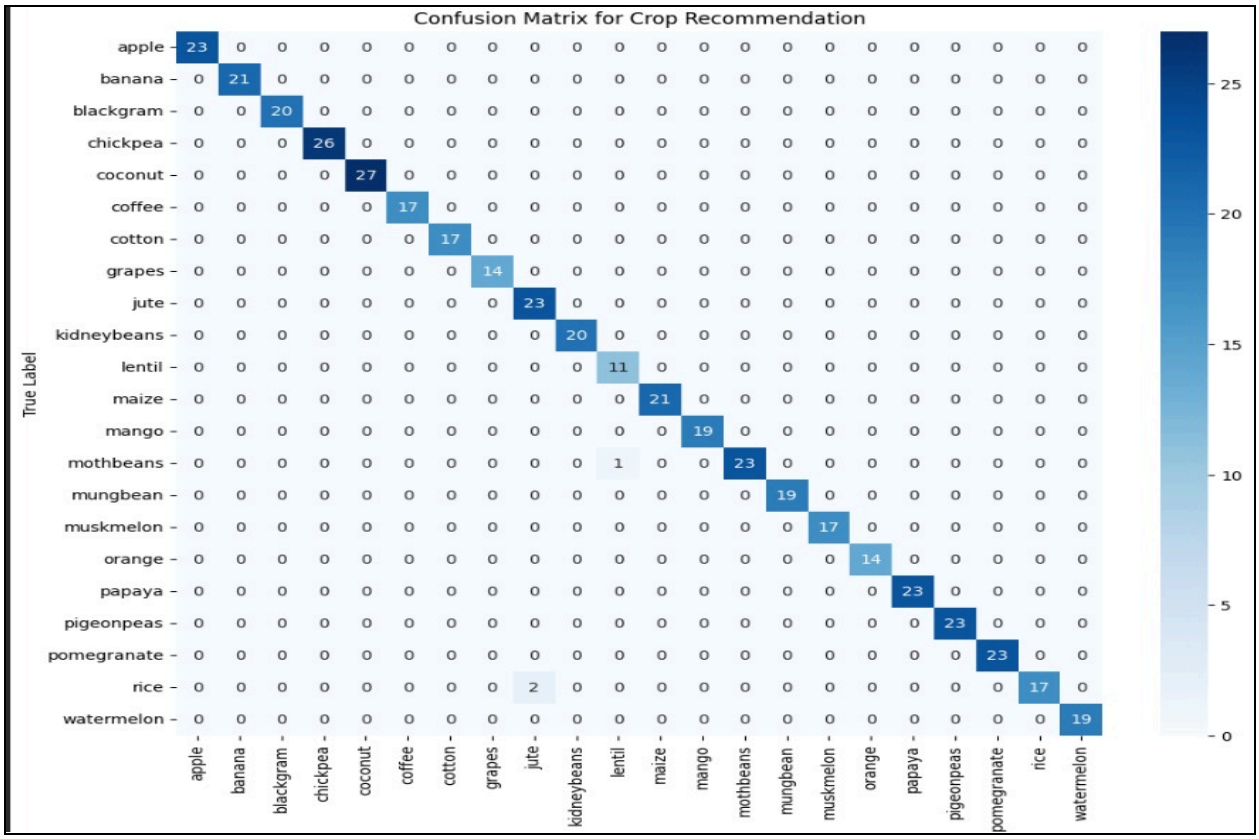


Fig 5 : Confusion matrix for Crop Recommendation

The above figure shows a Confusion Matrix for Crop Recommendation, which evaluates the performance of a machine learning model used in the KrishiMitra system. Each row represents the true crop label, while each column represents the predicted crop. The diagonal values indicate correct predictions — showing that most crops like apple (23), banana (21), chickpea (26), coconut (27), jute (23), and others were accurately classified. The matrix is highly diagonal, meaning the model performs exceptionally well with minimal misclassifications. The color intensity from light to dark blue indicates prediction strength, with darker shades representing higher accuracy. Overall, this confusion matrix reflects a highly accurate crop recommendation .

5.2 Plant Disease Accuracy

The model was evaluated using a 10-fold cross-validation approach to ensure robust performance assessment. The fold-wise accuracies remained consistently high, ranging from 0.9909 to 1.0000,

indicating that the model maintained stability across different data splits. The calculated mean accuracy was 0.9959, with minimal standard deviation, confirming the model's reliability and strong generalization capability. During the training phase, the PlantCNN model was optimized using the Adam optimizer and CrossEntropyLoss over five epochs. The training loss progressively decreased from 1.35 in the first epoch to 0.26 in the fifth, demonstrating effective learning and convergence. Finally, the trained model achieved a validation accuracy of 87.53%, reflecting its efficiency in correctly identifying and classifying plant images from unseen data.

5.3 Market Analysis

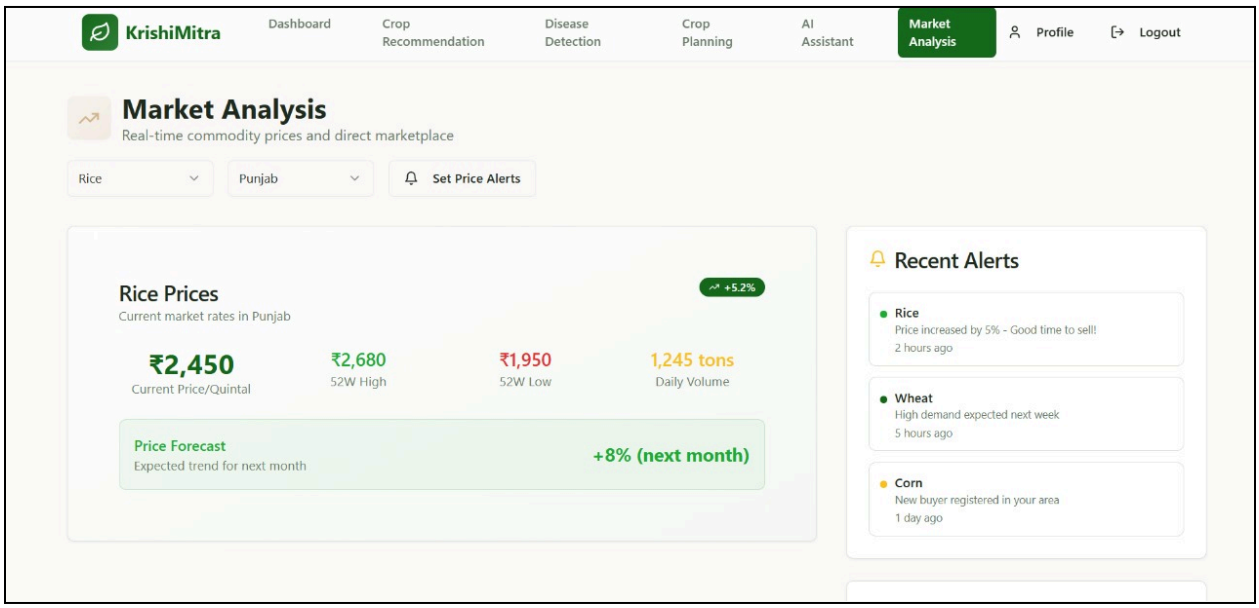


Fig 6 : Market Analysis Dashboard

The above figure shows the Market Analysis dashboard of the KrishiMitra application, which provides real-time commodity prices and market insights. It displays current rice prices in Punjab ₹2,450 per quintal, with a 52-week high of ₹2,680 and a low of ₹1,950. The price forecast predicts an 8% increase next month. A section on the right lists recent alerts such as a 5% rice price hike, high wheat demand, and a new corn buyer notification. The interface is clean, informative, and farmer-friendly.

5.4 AI Agricultural Assistant

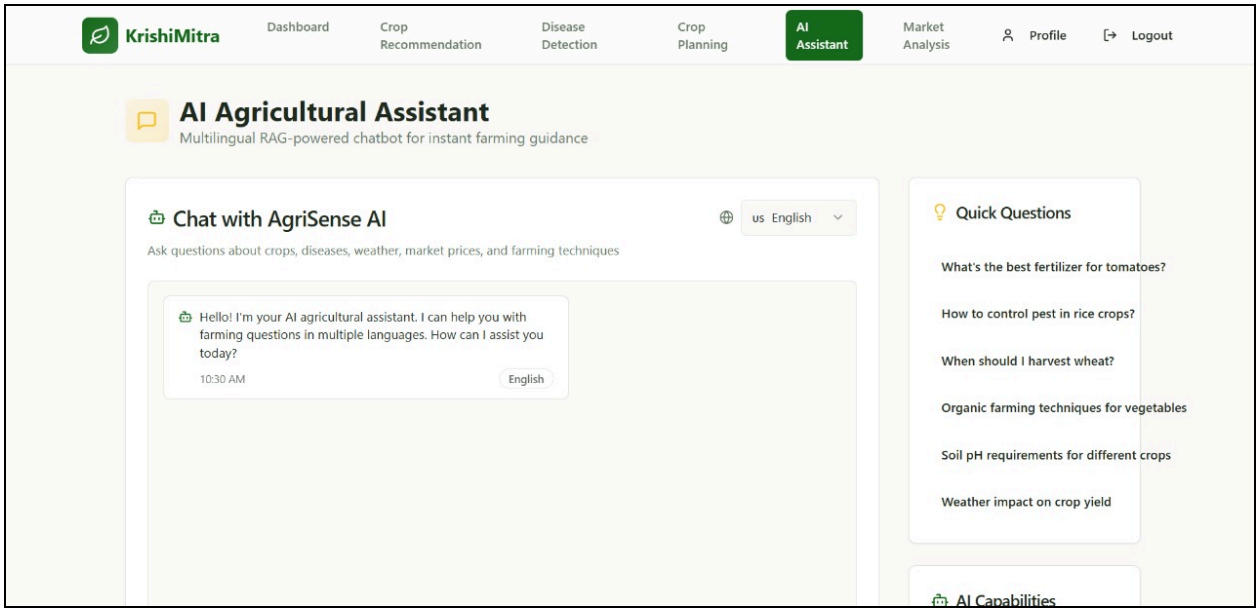


Fig 7: Chatbot Dashboard

The above figure shows the AI Agricultural Assistant page of KrishiMitra, featuring the AgriSense AI chatbot that provides multilingual farming guidance. Users can ask questions about crops, weather, and market prices. The interface includes a chat window, a language selector, and a Quick Questions panel with common farming queries like fertilizer use, pest control, and harvesting tips.

Chapter 6. Plan of action for the next semester

6.1 Work Done Till Date

So far, the **KrishiMitra** system has been successfully designed and partially implemented with core functionalities. The completed work includes the development of modules for Crop Recommendation using the Random Forest algorithm, Disease Detection using CNN, AI-Based Weekly Crop Plan Generation, RAG-Based Chatbot integration, and Market Analysis. The frontend has been designed using React.js and Tailwind CSS, while the backend is supported by Flask and Node.js. The system currently provides accurate crop insights, disease diagnosis, and real-time market trends through a user-friendly interface.

6.2 Plan of Action for Project II

In the next semester, we plan to enhance KrishiMitra by integrating a Voice Assistant into the existing chatbot, enabling farmers to interact using speech commands instead of text. Additionally, Multilingual Support will be implemented so users can access the system in their preferred local languages. These improvements will make the platform more accessible, interactive, and inclusive, helping farmers engage with KrishiMitra more naturally and effectively in their daily agricultural tasks.

Chapter 7.Conclusion

The project successfully integrates artificial intelligence, machine learning, and data-driven analytics to support farmers in making smarter agricultural decisions. By combining modules for

crop recommendation, plant disease detection, AI-based weekly crop planning, chatbot-based assistance, and market analysis, the system offers a comprehensive digital solution for modern farming challenges. It empowers farmers with real-time insights, personalized recommendations, and easy access to agricultural knowledge in multiple languages. Overall, KrishiMitra bridges the gap between technology and agriculture, promoting sustainable farming practices and enhancing productivity through intelligent automation.

Chapter 8. References

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Chapter 9. Appendix

a.List of Figures

Figure No.	Heading	Page no

1	Block Diagram	22
2	DFD(lvl 0)	23
3	DFD(lvl1)	23
4	Gantt Chart	25
5	Confusion Matrix	26
6	Market Analysis	27
7	ChatBot	28

b. List of Tables

Table No	Heading	Page no.
1	Comparison of existing systems and proposed area of work	17

c. Review 1 Sheet and 2

51

Industry/Inhouse: Inhouse.
Project type: Application/Research/Product
SDG Type:

Project Evaluation Sheet 2025-26
Semester 7

Class: D17 A/B/C

Title of Project(Group no): Krishimitra- (51)

Group Members: Jenny Lalwani (37) D17A, Vineet Chelani (17) D17A, Sahil Ahuja (03) D17A, Sahil Ahuja (3) D17B

	Engineering Concepts & Knowledge (5)	Investigation & Interpretation of Problem statement (5)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal & Environmental Benefit, Safety Consideration (2)	Ethics (2)	Cost effective-ness (2)	Team work (2)	Presentation Skills (3)	Engg. Mgmt principles & Standards Applied (3)	Life - long learning (3)	Professional Skills (5)	Innovative Approach (5)	Total Marks (50)
Review of Project Stage 1	4	3	3	2	4	2	2	2	2	2	3	3	3	3	38

Comments:

Name & Signature Reviewer1 Dr Sujata Khedkar

	Engineering Concepts & Knowledge (5)	Investigation & Interpretation of Problem statement (5)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal & Environmental Benefit, Safety Consideration (2)	Ethics (2)	Cost effective - ness (2)	Team work (2)	Presentation Skills (3)	Engg. Mgmt principles & Standards Applied (3)	Life - long learning (3)	Professional Skills (5)	Innovative Approach (5)	Total Mark (50)
Review of Project Stage 1	4	3	3	2	4	2	2	2	2	2	3	3	4	8	39

Comments:

Date: 19th September, 2025

Name & Signature Reviewer 2 Priti Joshi

51

Industry/Inhouse: Inhouse.
Project type: Application/Research/Product
SDG Type:

Project Evaluation Sheet 2025-26
Semester 7

Class: D17 A/B/C D17A.

Title of Project(Group no): KrishiMitri (51)

Group Members: Jenny Lalwani (D17A), Sahil Ahuja (D17B), Vineet Chelani (D17A), Sahil Ahuja (D17A). (Absent)

	Engineering Concepts & Knowledge (5)	Investigation & Interpretation of Problem statement (5)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal & Environmental Benefit, Safety Consideration (2)	Ethics (2)	Cost effective-ness (2)	Team work (2)	Presentation Skills (3)	Engg. Mgmt principles & Standards Applied (3)	Life - long learning (3)	Professional Skills (5)	Innovative Approach (5)	Total Marks (50)
Review of Project Stage 1	4	4	4	3	3	2	2	2	2	3	2	3	5	5	44

Comments: create more integrated form of weekly planner, also make in the form of video.

Name & Signature Reviewer1 Dr. Sujata Khedkar

	Engineering Concepts & Knowledge (5)	Investigation & Interpretation of Problem statement (5)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal & Environmental Benefit, Safety Consideration (2)	Ethics (2)	Cost effective - ness (2)	Team work (2)	Presentation Skills (3)	Engg. Mgmt principles & Standards Applied (3)	Life - long learning (3)	Professional Skills (5)	Innovative Approach (5)	Total Marks (50)
Review of Project Stage 1	4	4	4	3	3	2	2	2	2	3	2	3	5	5	44

Comments: With Disease detection also implement improvement model.

Date: 17th October, 2025

Name & Signature Reviewer 2 Priti Joshi